

Identifying Risk-Neutral Densities from an Empirical Measure of Listed Quotes

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Abstract

European option prices of a single maturity encode the risk-neutral density of the underlying at expiry. Differentiating those prices twice is ill-posed on a finite strike list. The calls are rescaled to a complementary distribution, the empirical measure of the risk-neutral jumps in strike, and that measure is fit by a smoothing spline. A Gaussian damping of its characteristic function inverts in closed form and, after the unquoted tails are filled and the kernel is thriced, supplies both the density and the call. The penalty is zero when the thriced density is already unimodal. The estimator is in-fill consistent. On four listed chains the out-of-the-money error is below the positive convolution approximation, each fitted density has one peak and mass 1.0, and on noisy Heston quotes the mean integrated squared error is below that convolution as well.

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1. Introduction

Breeden and Litzenberger (1978) identified the risk-neutral density of the underlying at a fixed expiry with the second strike derivative of the corresponding European call, scaled by a discount factor. The identity assumes a twice differentiable price curve. A market supplies a finite ordered list of strikes and a

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quote at each knot. Numerical second differentiation of that list is ill-posed: interpolants oscillate, change sign, and depend on the interpolating scheme. The literature responds with parametric mixtures and Gram–Charlier expansions, implied-volatility interpolants, local polynomials of the call, and Bondarenko (2003)’s positive convolution approximation, which forces the density to be a mixture of a positive kernel with a discrete probability fitted to prices. Section 2 reviews these approaches.

This paper estimates that density from the empirical measure of a single maturity. The calls are rescaled to a complementary distribution, which records the risk-neutral jumps in strike. A smoothing spline fits the measure: a natural cubic, with a penalty on the second derivative that is zero when the thrice density is already unimodal and positive when that derivative is noisy. A Gaussian damping of the characteristic function of the spline inverts in closed form. The same smoothing is the density and, after one integration, the call. Tails complete the unquoted wings, and thricing raises the order of the kernel. With those pieces in place, the estimator converges to the true density on the interior of a compact interval as the strike mesh fills that interval and the bandwidth tends to zero, and the same limit holds for noisy quotes when $nh^5 \rightarrow \infty$. The exact-quote argument is proved for the interpolating spline. On the exact designs below the selected penalty is zero, so the reported fit is that spline. The noise argument extends to a penalty chosen from any fixed finite grid of c , which is the rule of Section 4.3.

We compare the fit to several extant methods. Bondarenko’s method convolves a positive kernel with a discrete probability, so the estimate is a density by construction. Aït-Sahalia and Duarte (2003) smooth a decreasing-convex projection of the calls with a local polynomial, so the density is nonnegative by design and the call cannot match quotes that violate convexity. Aït-Sahalia and Lo (1998) smooth implied volatility, and Grith et al. (2012) use a local cubic of the call and of implied volatility. On the listed chains of Section 7 the out-of-the-money root-mean-square error (RMSE) is below the convolution of Bondarenko, and on noisy Heston quotes the mean integrated squared error is below it as well.

The remainder of this manuscript is structured as follows. Section 2 reviews related work. Section 3 constructs the estimator. The characteristic function of the normalized call supplies the closed form. Section 4 records three refinements: completing the tails, thricing, and the penalty. Section 5 proves in-fill consistency for exact quotes and for noisy quotes. Section 6 chooses bandwidths that meet those conditions and compares the estimator with those of Section 7 on exact Heston and variance-gamma quotes and on noisy Heston quotes. Section 7 presents an empirical illustration on four listed chains. Section 8 concludes.

2. Related Work

The identity of Section 3 is the continuous-strike version of a spanning argument of Ross (1976): if one could trade a put at every strike, the second derivative of the put curve would be a density, and one could synthesize any sufficiently regular payoff. Dupire (1994) obtained the analogous result in calendar time, which produces a local-volatility equation. Observed markets supply neither a continuum of strikes nor noiseless quotes.

A first family of methods specifies the density of S_T as a parametric curve and fits the implied option prices to the quotes. Ritchey (1990) and Melick and Thomas (1997) take the density to be a mixture of lognormals, so that the call price is a mixture of Black–Scholes prices; Bahra (1997) and Söderlind and Svensson (1997) use related specifications for interest-rate and foreign-exchange options. Two or three components typically suffice to generate a unimodal, skewed density, but the fit is poorly identified once more than two components are admitted, and the approximation to a Heston or jump-diffusion density need not vanish as more quotes arrive. Expansion methods perturb a baseline density—usually lognormal—by Gram–Charlier or Hermite terms (Jarrow and Rudd, 1982; Corrado and Su, 1996; Rompolis and Tzavalis, 2007; Xiu, 2014). The expansions can match a finite number of moments, but they are not guaranteed to stay nonnegative, and the truncated series can oscillate in the tails.

A second, and in applied work the most common, family interpolates the Black–Scholes implied-volatility function of strike and converts the interpolant back to prices before differentiating twice. Shimko (1993) fitted a quadratic in strike; Malz (1997) used a cubic spline in delta; Bliss and Panigirtzoglou (2002) a smoothing spline. Figlewski (2010) completes the tails with a generalized extreme-value distribution. Because implied volatility is a much flatter function of strike than the call itself, a low-order interpolant often has small bias when quotes are accurate. The drawback is that implied volatility is a nonlinear transformation that inflates price noise where the derivative of price with respect to implied volatility is small, i.e. in the tails. An interpolating spline then treats those noisy tail quotes as exact, and the second derivative of the resulting call curve oscillates and can become negative. Shape-constrained smoothers of the implied-volatility surface (Fengler, 2005, 2009; Gatheral and Jacquier, 2014; Ludwig, 2015) address the sign problem at the cost of a more involved optimization.

A third family estimates the call, or its Black–Scholes implied volatility, by kernel regression or local polynomials and reads the density off a second strike

derivative. Aït-Sahalia and Lo (1998) use a Nadaraya–Watson kernel. The state-price densities in that paper are estimated from 14,431 S&P 500 options pooled across every trading day of 1993, because a single expiry was treated as too short a list on which to differentiate twice. They emphasise that the bandwidth dominates the choice of kernel: too small a neighborhood reproduces the noise in the quotes, and too large a neighborhood removes the curvature of the smile. Aït-Sahalia and Lo (2000) used the same estimator to recover an implied risk-aversion function. Aït-Sahalia and Duarte (2003) work one day at a time, on the twenty to fifty strikes of a single expiry, and impose monotonicity and convexity of C in K . The local polynomial that the asymptotics recommend for a second derivative is a local cubic (Fan and Gijbels, 1996). On 242 days in 1999, with the 25 most active strikes, unconstrained Nadaraya–Watson, local linear, local quadratic, and local cubic estimates of the density took a negative value on 242, 130, 205, and 241 days. The shape-constrained local linear never did. Stopping the violations in the unconstrained fits required substantial oversmoothing and a large bias. The raw quotes themselves broke convexity on 24% of the high-frequency prints that year.

Grith et al. (2012) survey this literature and estimate a local cubic of the rescaled call with a quartic kernel. For a second derivative the optimal bandwidth is of order $n^{-1/9}$. On their one-month DAX chain, least-squares cross-validation selects $h = 114$, which they describe as wiggly, so the densities they plot use $h = 228$ and $h = 434$. The implied-volatility smoother they display is widened further, to $h = 372$. Smoothing the call is more sensitive to that choice than smoothing implied volatility, and the tails oscillate unless the bandwidth is increased at a cost in bias. Twice differentiating a Nadaraya–Watson or Gasser and Müller (1979) level fit is set aside as the statistically worse route to the same derivative. Jackwerth (2004) starts a kernel regression of implied volatility from Silverman’s rule, of order $n^{-1/5}$. He warns that the bandwidth which prices overfits noise from one day to the next, that a kernel does not continue across gaps in the strike list, and that a spline has to be of degree higher than three if the recovered distribution is to be smooth. Figlewski (2018) finds that filling the implied-volatility curve with a cubic spline puts spikes in the density, because the second derivative of a cubic is continuous but not differentiable, and recommends a spline of degree four or higher. Fan and Mancini (2009) combine a parametric guide with a nonparametric correction. Related shape-constrained least squares are studied by Yatchew and Härdle (2006) and Birke and Pilz (2009).

The estimator of Section 3 is a kernel smooth of that empirical measure on a single maturity. The pilot is a smoothing spline of the complementary distribution, the kernel is Gaussian, and the bandwidth is a fixed multiple of $n^{-1/9}$. The penalty

of Section 4.3 is what responds when the second derivative is noisy. Thricing raises the order of the kernel and puts negative sidelobes in the equivalent kernel for $f_{\mathbb{Q}}$. The convolution is closed form. Section 7 implements the Nadaraya–Watson implied-volatility regression of Aït-Sahalia and Lo at the bandwidth in their Appendix A, the local cubics of Grith et al. at the leave-one-out width and at twice that width, and the two-step procedure of Aït-Sahalia and Duarte: the same projection of the quoted calls, followed by one Gaussian local linear at the Fan–Gijbels plug-in. The density is the scaled and shifted derivative of that local linear.

Bondarenko (2003) proposed the positive convolution approximation (PCA). The density is required to lie in the set of functions obtained by convolving a discrete probability with a positive kernel, and the discrete probability is chosen to minimize the sum of squared pricing errors. The kernel enforces smoothness and, together with positive weights, guarantees that the result is a density. In Bondarenko’s comparison, the method dominated mixtures of lognormals, Hermite expansions, and several regularization methods. Subsequent simulation evidence on affine jump-diffusions is consistent with this ranking (Lu, 2019). Section 7 implements that convolution with a Gaussian kernel on the spot. The mixing nodes are equally spaced from the first quoted strike to the last, the weights are a probability with mean F , and the bandwidth is chosen by even/odd pricing error.

Rubinstein (1994) and Jackwerth and Rubinstein (1996) recovered probabilities on a binomial tree by minimizing a distance to a prior subject to pricing constraints. Stutzer (1996) and Buchen and Kelly (1996) maximized entropy. These methods produce a valid density by construction, but the finite-support discretization and the choice of prior affect the tails. Viewing the problem as a linear inverse problem, Cont and Tankov (2004) calibrated jump-diffusion models under a relative-entropy penalty; Monnier (2014) used the singular value decomposition of a restricted put operator.

A number of papers compare methods on market data, where the true density is unknown, so that the criterion is typically pricing error or stability over time (Cooper, 1999; Jondeau and Rockinger, 2000; Coutant et al., 2001; Anagnou et al., 2005; Bliss and Panigirtzoglou, 2002; Liu et al., 2007). Studies that generate options from a known process are less common. Bondarenko (2003) finds that his method outperforms mixtures, expansions, and several regularization methods on integrated squared error. Lai (2014) finds that kernel regression outperforms spline interpolation and neural nets on pricing error. Santos and Guerra (2015) compare mixtures, Edgeworth expansions, hypergeometric functions, and a Bliss and Panigirtzoglou smoothing spline against Heston and other

targets. Lu (2019) compares methods on affine jump-diffusions and again finds Bondarenko's method accurate. The typical ranking is that implied-volatility interpolants price well when quotes are clean, that mixtures are unstable once more than two components are admitted, and that Bondarenko's method is the most accurate among methods that do not use a parametric density of S_T , when the true density is known. Section 6.1 is in this vein: the targets are Heston and variance-gamma, and the true density is known. Section 7 scores listed index chains, where the criterion is pricing error, against Bondarenko, Aït-Sahalia and Duarte, Aït-Sahalia and Lo, and Grith et al..

3. Construction of the Estimator

3.1. Construction

Assume a risk-neutral measure $\mathbb{Q}$ under which S_T has a density $f_{\mathbb{Q}}$ on $(0, \infty)$ and $\mathbb{E}^{\mathbb{Q}}[S_T] < \infty$. Let r be the continuously compounded interest rate and q the continuous dividend yield. Write $F = S_0 e^{(r-q)T}$ for the forward. Under $\mathbb{Q}$,

$$\begin{aligned} S_0 e^{-qT} &= e^{-rT} \mathbb{E}^{\mathbb{Q}}[S_T] \\ &= e^{-rT} \int_0^{\infty} x f_{\mathbb{Q}}(x) dx, \end{aligned} \tag{1}$$

hence $\mathbb{E}^{\mathbb{Q}}[S_T] = F$. Write C for the corresponding call curve:

$$\begin{aligned} C(K) &= e^{-rT} \mathbb{E}^{\mathbb{Q}}[(S_T - K)^+] \\ &= e^{-rT} \int_K^{\infty} (x - K) f_{\mathbb{Q}}(x) dx. \end{aligned}$$

Now define the normalized option price

$$\tilde{C}(K) \equiv \frac{C(K)}{S_0 e^{-qT}} = \frac{\int_K^{\infty} (x - K) f_{\mathbb{Q}}(x) dx}{F}. \tag{2}$$

The normalized call price is a decreasing function of K : it is equal to 1 if $K = 0$ and equal to 0 if $K = \infty$. Thus the normalized call price, as a function of K , is a complementary cdf, say $\tilde{C}(K) = 1 - P(K)$. The characteristic function (hereafter c.f.) of the law of P is the Fourier–Stieltjes transform

$$\varphi_P(s) = \int_{(0, \infty)} e^{isK} dP(K). \tag{3}$$

For a cdf on $[0, \infty)$ and $s \neq 0$, (3) is

$$\varphi_P(s) = 1 + is \int_0^\infty e^{isK} \tilde{C}(K) dK.$$

Substitute (2). Fubini yields

$$\int_0^\infty e^{isK} \tilde{C}(K) dK = \frac{1}{F} \int_0^\infty f_{\mathbb{Q}}(x) \left[\int_0^x e^{isK} (x - K) dK \right] dx.$$

The inner integral is $-x/(is) + (e^{isx} - 1)/(is)^2$. Hence, for $s \neq 0$,

$$\begin{aligned} \varphi_P(s) &= 1 + \frac{is}{F} \int_0^\infty f_{\mathbb{Q}}(x) \left[-\frac{x}{is} + \frac{e^{isx} - 1}{(is)^2} \right] dx \\ &= 1 - F^{-1} \mathbb{E}^{\mathbb{Q}}[S_T] + \frac{\varphi_{\mathbb{Q}}(s) - 1}{is F} \\ &= \frac{\varphi_{\mathbb{Q}}(s) - 1}{is F}, \end{aligned} \tag{4}$$

where $\varphi_{\mathbb{Q}}(s) = \mathbb{E}^{\mathbb{Q}}[e^{isS_T}]$ and $\mathbb{E}^{\mathbb{Q}}[S_T] = F$ by (1). Solving for $\varphi_{\mathbb{Q}}$, one obtains

$$\varphi_{\mathbb{Q}}(s) = 1 + is F \varphi_P(s). \tag{5}$$

Read from the right, (4) is a pricing formula: a model that supplies $\varphi_{\mathbb{Q}}$ yields φ_P , hence the complementary cdf $\tilde{C} = 1 - P$ and the call (Lewis, 2000). This paper inverts the identity: φ_P is constructed from the quotes, and $\varphi_{\mathbb{Q}}$ is recovered.

The c.f. $\varphi_{\mathbb{Q}}$ is that of S_T . Fourier inversion yields

$$f_{\mathbb{Q}}(K) = \frac{1}{2\pi} \int_{-\infty}^\infty e^{-isK} \varphi_{\mathbb{Q}}(s) ds. \tag{6}$$

Substitute (5) into (6):

$$f_{\mathbb{Q}}(K) = \frac{1}{2\pi} \int_{-\infty}^\infty e^{-isK} ds + \frac{F}{2\pi} \int_{-\infty}^\infty e^{-isK} (is) \varphi_P(s) ds.$$

The first integral is a Dirac mass at $K = 0$. For $K > 0$,

$$f_{\mathbb{Q}}(K) = \frac{F}{2\pi} \int_{-\infty}^\infty e^{-isK} (is) \varphi_P(s) ds. \tag{7}$$

Observed calls at strikes $0 < K_1 < \dots < K_m$ supply values $\tilde{C}_i = C_i/(S_0 e^{-qT})$ and hence $P_i = 1 - \tilde{C}_i$. The estimator interpolates these knots by a natural cubic

spline P^{cs} . The spline matches value and first derivative at the knots. The c.f. of this absolutely continuous measure is $\int e^{isK} P^{cs'}(K) dK$. Replace φ_P by the product of that integral with the Fourier transform of a Gaussian kernel of width h ,

$$\hat{\varphi}_P(s) = \left(\int_{(0,\infty)} e^{isK} P^{cs'}(K) dK \right) e^{-h^2 s^2 / 2}. \quad (8)$$

The Gaussian density and its distribution function are

$$\kappa_h(u) = \frac{1}{h\sqrt{2\pi}} \exp\left(-\frac{u^2}{2h^2}\right), \quad \mathcal{K}_h(u) = \int_{-\infty}^u \kappa_h.$$

The factor $e^{-h^2 s^2 / 2}$ is convolution against κ_h , and multiplication by is is differentiation up to sign. Substituting (8) into (7) therefore yields

$$\hat{f}_{\mathbb{Q}}(K) = -F(P^{cs'} * \kappa_h)'(K), \quad \hat{P}(K) = (P^{cs'} * \mathcal{K}_h)(K). \quad (9)$$

On each cell the integrand is a quadratic times a Gaussian. The integrals of 1, u , and u^2 against ϕ and Φ are elementary, so (9) is closed form. Calls are $\hat{C} = S_0 e^{-qT} (1 - \hat{P})$. These knots are the quoted strikes. Section 4.1 adds the endpoints that complete the distribution.

4. Two Refinements

4.1. Completing the Tails

The spline of Section 3.1 stops at the first and last quotes. On $(0, K_1)$ it does not yet specify P , and if $C_m > 0$ the residual mass $1 - P(K_m)$ is unplaced. Both ends are wrong as a cdf on $(0, \infty)$.

On $(0, K_1]$ a deep in-the-money call is intrinsic, $C(K) = e^{-rT}(F - K)$, hence $\tilde{C}(K) = 1 - K/F$ and $P(K) = K/F$. That is a straight line through the origin. The estimator matches the observed $P(K_1)$ rather than K_1/F ,

$$P(K) = P(K_1) \frac{K}{K_1}, \quad 0 < K \leq K_1. \quad (10)$$

When the first quote is at intrinsic, $P(K_1) = K_1/F$ and (10) is exactly K/F . The spline imposes the endpoint $P(0) = 0$, which is the same line at the boundary.

On the right, $\tilde{C}(\infty) = 0$ so $P(\infty) = 1$. If $C_m > 0$, a European call still has to fall to zero. The residual call is continued linearly with slope in $[-e^{-rT}, 0)$ until $C = 0$ at an intercept K_{end} . The slope is the last strictly negative step of the quoted

call, clipped to that interval. A flat last step, two quotes on the premium floor, is not a decay: using it would send K_{end} to the cap $8F$. The spline imposes the single endpoint $P(K_{\text{end}}) = 1$. The bandwidth still counts a notional knot spacing along that line, eight per last quoted gap,

$$n_R = \text{round}(8(K_{\text{end}} - K_m)/(K_m - K_{m-1})), \quad (11)$$

are equally spaced in $(K_m, K_{\text{end}}]$ and converted to $P = 1 - \tilde{C}$. If $C_m = 0$, $P(K_m) = 1$ already and this wing is empty.

Only the left completion changes the estimator on the Heston and variance-gamma chains below: the first quoted call is at intrinsic, and the last quoted call is already zero (dense) or of order 10^{-5} (sparse).

4.2. Thricing

The Gaussian kernel κ_h has second moment h^2 , so the leading bias of (9) is $O(h^2)$. When the quotes are treated as exact there is no observation-error variance to inflate, and a higher-order kernel is admissible. Twicing (Schucany and Sommers, 1977) replaces $e^{-h^2 s^2/2}$ by $2e^{-h^2 s^2/2} - e^{-h^2 s^2}$, the transform of $2\kappa_h - \kappa_{h\sqrt{2}}$. That kernel has a vanishing second moment, and the bias falls from $O(h^2)$ to $O(h^4)$. One further bandwidth cancels the new leading term. The transform in (8) is replaced by

$$\frac{8}{3}e^{-h^2 s^2/2} - 2e^{-h^2 s^2} + \frac{1}{3}e^{-2h^2 s^2}, \quad (12)$$

which is the Fourier transform of $\frac{8}{3}\kappa_h - 2\kappa_{h\sqrt{2}} + \frac{1}{3}\kappa_{2h}$. The moments of order 2 and 4 vanish, so the bias is $O(h^6)$ when $f_{\mathbb{Q}}$ has a bounded sixth derivative. In closed form,

$$\hat{f}_{\mathbb{Q}}^{\text{th}}(K) = \frac{8}{3}\hat{f}_{\mathbb{Q},h}(K) - 2\hat{f}_{\mathbb{Q},h\sqrt{2}}(K) + \frac{1}{3}\hat{f}_{\mathbb{Q},2h}(K), \quad (13)$$

and the same combination is applied to the interpolant in (9). The equivalent kernel takes negative values, so $\hat{f}_{\mathbb{Q}}^{\text{th}}$ is not a density. The call is $S_0 e^{-qT}(1 - \hat{P}^{\text{th}})$, with $\hat{P}^{\text{th}}$ clipped to $[0, 1]$ and both premiums floored at zero.

Thricing a truncated distribution amplifies leakage from the missing tails, so it is applied only after Section 4.1.

4.3. The penalty

The spline in (9) interpolates every quoted value of P . When those values are exact, that is the right constraint. When the quotes are noisy, the second derivative

that thricing amplifies is not. The estimator replaces the interpolant by the natural cubic f that minimises

$$\sum_i (P(K_i) - f(K_i))^2 + c F^3 \int (f''(K))^2 dK, \quad (14)$$

with the tail knots of Section 4.1 included among the K_i . The factor F^3 makes c dimensionless. At $c = 0$ the minimiser is the interpolant, and (13) is the estimator of the preceding subsection. The solution remains a cubic spline, so the convolution stays closed form.

The candidates are $c = 0$ and sixteen values spaced geometrically from 10^{-7} to 3×10^{-4} . Let c_0 be the smallest candidate at which $\hat{f}_{\mathbb{Q}}^{\text{th}}$ has one peak on $[0.55F, 1.40F]$ and total variation at most 1.02. A curve that already meets that test keeps c_0 . Above c_0 , the log root-mean-square gap between the fitted calls and the input quotes is plotted against the log root-mean-square gap between $\hat{f}_{\mathbb{Q}}^{\text{th}}$ and the plain Gaussian convolution, on that same window. If the discrete curvature of the curve is at least $1/2$, the penalty is the corner. A flatter bend keeps c_0 .

The estimator used below is (14) with the tails, followed by (13) at the bandwidth (16). On the exact Heston and variance-gamma quotes the selected penalty is zero.

5. In-Fill Consistency

Write P for the complementary-call cdf of Section 3, so that $f_{\mathbb{Q}} = -F P''$ on $(0, \infty)$. The estimator (9) replaces P' by the derivative of its cubic spline and smooths with κ_h .

Theorem 1. *Let $\mathbb{E}^{\mathbb{Q}}[S_T] < \infty$, and let $f_{\mathbb{Q}}$ be four times continuously differentiable on $(0, \infty)$. Let C be the corresponding call curve. Fix a compact interval $[A, B] \subset (0, \infty)$. Consider a sequence of exact quote grids $0 < K_1^{(m)} < \dots < K_m^{(m)}$ with mesh $\delta_m = \max_i (K_i^{(m)} - K_{i-1}^{(m)}) \rightarrow 0$, $K_1^{(m)} \leq A$ and $K_m^{(m)} \geq B$. Form the cubic spline of Section 3.1 and complete the tails as in Section 4.1. Let $h_m \rightarrow 0$. Then both (9) and its thriced version (13) converge to $f_{\mathbb{Q}}$ uniformly on every compact subset of (A, B) .*

Proof. The estimator is $\hat{f}_{\mathbb{Q}} = -F(P^{\text{CS}'} * \kappa_h)'$, and $f_{\mathbb{Q}} = -FP''$. Split

$$(P^{\text{CS}'} * \kappa_h)' - P'' = ((P^{\text{CS}'} - P') * \kappa_h)' + (P' * \kappa_h - P')'.$$

The second term is ordinary kernel bias of $P'' = -f_{\mathbb{Q}}/F$. Because κ is even, $\int \kappa = 1$ and $\int u\kappa(u) du = 0$. If $f''_{\mathbb{Q}}$ is bounded on a neighborhood of $[A, B]$, the difference is $O(h^2)$ uniformly on compact subsets of (A, B) several bandwidths away from the endpoints. The thrice kernel $\frac{8}{3}\kappa_h - 2\kappa_{h\sqrt{2}} + \frac{1}{3}\kappa_{2h}$ has Fourier transform (12). Its moments of order 2 and 4 vanish, so the same expansion starts at order h^6 whenever $f_{\mathbb{Q}}$ is C^6 .

The first term is interpolation. On a quasi-uniform mesh, the natural cubic spline of a C^4 function converges to the function, and its first derivative converges to the first derivative, uniformly on compact subsets of the interior of the grid. Thus $P^{\text{cs}'} \rightarrow P'$ as $\delta_m \rightarrow 0$. Convolution against κ_h preserves the limit, and one derivative in K does not change the conclusion on those compacts.

The filled wings are supported on $(0, K_1]$ and, if $C_m > 0$, on $(K_m, K_{\text{end}}]$. Their contribution at an evaluation point $K \in [A + \varepsilon, B - \varepsilon]$ is at most a constant times $h^{-2} \exp(-\varepsilon^2/2h^2)$, because $K_1 \leq A$ and $K_m \geq B$. That bound tends to zero faster than any power of h .

Multiplying by $-F$ yields the claim for $\hat{f}_{\mathbb{Q}}$ and for $\hat{f}_{\mathbb{Q}}^{\text{th}}$. $\square$

Theorem 2. *Keep the assumptions of Theorem 1 on $f_{\mathbb{Q}}$ and on the grids, and assume in addition that the mesh on $[A, B]$ is quasi-uniform: $n_m\delta_m$ is bounded above and below, where n_m is the number of strikes in $[A, B]$. Observe*

$$P_i^{(m)} = P(K_i^{(m)}) + \eta_i^{(m)},$$

with $\eta_i^{(m)}$ independent, $\mathbb{E}[\eta_i^{(m)}] = 0$ and $\mathbb{E}[(\eta_i^{(m)})^2] \leq \tau^2 < \infty$. Form the cubic spline and the tails from these noisy knots. Let $h_m \rightarrow 0$ and $n_m h_m^5 \rightarrow \infty$. Then both (9) and (13) converge in probability to $f_{\mathbb{Q}}$, uniformly on every compact subset of (A, B) .

Proof. The spline operator is linear, so the spline of the noisy knots is the spline of the true knots plus the spline S_{η} of $(K_i^{(m)}, \eta_i^{(m)})$, up to the tail knots. The tail knots lie outside $[A, B]$. Their contribution on a compact several bandwidths inside (A, B) is of order $h^{-2} \exp(-c/h^2)$ as in the proof of Theorem 1, whether or not the intercept K_{end} depends on the last noisy slope, because that intercept stays inside a fixed multiple of F .

The piece built from the true knots converges to $f_{\mathbb{Q}}$ by Theorem 1. The noise piece is $-F(S'_{\eta} * \kappa_h)'$. On a quasi-uniform mesh, once $h/\delta \rightarrow \infty$, this is within $o_p(1)$ of the kernel estimator

$$\sum_{i=1}^{n_m} \eta_i^{(m)} \delta_i h_m^{-3} L((K - K_i^{(m)})/h_m),$$

where δ_i is the local gap and L is a fixed linear combination of κ'' (of $\frac{8}{3}\kappa''(\cdot) - 2\kappa''(\cdot/\sqrt{2}) + \frac{1}{3}\kappa''(\cdot/2)$ after thricing). The kernel L is bounded and integrable, and $\int L = 0$. The sum has conditional mean zero. Its conditional variance is of order

$$\tau^2 \sum_i \delta_i^2 h_m^{-6} L^2((K - K_i)/h_m) \asymp \tau^2 \delta_m h_m^{-5} \int L^2 = O(\tau^2/(n_m h_m^5)),$$

uniformly for K in a compact subset of (A, B) . The hypothesis $n_m h_m^5 \rightarrow \infty$ sends this variance to zero, and it also forces $h_m/\delta_m = n_m h_m \rightarrow \infty$. Chebyshev's inequality gives pointwise convergence in probability. The map $K \mapsto (S'_\eta * \kappa_h)'(K)$ is Lipschitz with a random constant of that same stochastic order, so a grid of spacing h_m upgrades the convergence to uniform convergence on the compact.

Thricing replaces L by a fixed linear combination of three such kernels and does not change the order of the variance. The bias expansion remains the one in the proof of Theorem 1. $\square$

Theorem 3. *Keep the assumptions of Theorem 2. Let the penalty c in (14) be any random variable taking values in a fixed finite set $C \subset [0, \infty)$, and let F stay fixed along the sequence. Form the smoothing spline, the tails, and (13) from the noisy knots. Then $\hat{f}_\mathbb{Q}^{\text{th}}$ converges in probability to $f_\mathbb{Q}$, uniformly on every compact subset of (A, B) .*

Proof. A maximum over the finite set C of terms that converge to zero in probability still converges to zero. It is therefore enough to fix $c \in C$ and to write $\lambda = cF^3$, a constant. At $c = 0$ the claim is Theorem 2. The smoothing spline is linear in the knot values for this fixed λ , so the fitted function is the spline of the true knots plus the spline of $(\eta_i^{(m)})$.

On the true knots, let P^{cs} be the interpolant and P_λ the smoothing spline. The variational inequality gives

$$\sum_i (P(K_i) - P_\lambda(K_i))^2 \leq \lambda \int (P^{\text{cs}''})^2,$$

and the integral stays bounded because P is C^4 and $P^{\text{cs}''}$ converges to P'' on the interior of the grid. The difference $P^{\text{cs}} - P_\lambda$ is the interpolating spline of these residuals. Once $h_m/\delta_m \rightarrow \infty$, its contribution to $(\cdot' * \kappa_{h_m})'$ is within $o(1)$ of

$$\sum_i r_i \delta_i h_m^{-3} L((K - K_i)/h_m).$$

Cauchy–Schwarz and the same counting as in Theorem 2 bound the sum by a constant times $(\lambda/(n_m h_m^5))^{1/2}$. The hypothesis $n_m h_m^5 \rightarrow \infty$ sends it to zero, uniformly on compact subsets of (A, B) . The interpolant P^{cs} is covered by Theorem 1, so the penalized spline of the true knots has the same limit.

On the noise, the Demmler–Reinsch basis diagonalizes both the empirical inner product and the penalty. The smoothing spline multiplies the coefficient of the j th basis function by $(1 + \lambda \rho_j)^{-1}$ with $\rho_j \geq 0$. Every linear functional of the fitted spline therefore has variance at most its variance at $\lambda = 0$. The map from the knots to $(S'_\eta * \kappa_{h_m})'(K)$ is such a functional, and at $\lambda = 0$ its variance is the $O(1/(n_m h_m^5))$ bound in the proof of Theorem 2. The noise piece is therefore $o_p(1)$, uniformly on those compacts, by the same Chebyshev and Lipschitz argument. Thricing again replaces the kernel by a fixed linear combination and does not change the order. $\square$

Remark 1. *Theorem 1 is a statement about exact quotes. Theorems 2 and 3 are the corresponding statements about additive noise in P . None of the three is a statement about the far tails outside $[A, B]$. The equivalent kernel of (13) takes negative values, so the thriced estimator is not necessarily a density. The empirical section fits the spline to the decreasing-convex projection of the quoted calls. When the true call lies in the interior of that cone, the projection changes the knots by an amount of the same stochastic order as η_i , and the bandwidth condition of Theorem 2 still applies.*

6. Choice of Bandwidth

The width h is the bandwidth in (13). Two rules are used below. Each is checked against Theorems 1 and 2. The penalty c is chosen from a fixed finite grid, and Theorem 3 gives the same limit for every choice on that grid.

With exact quotes the bias is the usual $O(h^2)$ kernel bias, or $O(h^6)$ after thricing, and the spline error vanishes with the mesh. For a Gaussian it is enough that h be a small multiple of the typical quoted gap. Write δ for the median of $\{K_i - K_{i-1}\}$ on the quoted strikes, not on the filled wings, and take

$$h = \min(1.2 \delta, 0.30 (K_m - K_1)). \quad (15)$$

The cap binds only on a very short chain. On an equally spaced grid every gap equals δ . Whenever $\delta \rightarrow 0$, this rule satisfies $h \rightarrow 0$, which is what Theorem 1 requires. It does not satisfy $nh^5 \rightarrow \infty$. A noisy chain with mesh δ and $h \propto \delta$ has nh^5 of order $\delta^4 \rightarrow 0$, so the variance in the proof of Theorem 2 explodes. The

exponent $n^{-1/5}$ familiar from kernel density estimation leaves the same product bounded, so it is not used.

The density is a second derivative. The bandwidth that balances an $O(h^2)$ bias against a variance of order $1/(nh^5)$ is of order $n^{-1/9}$, the rate Aït-Sahalia and Duarte (2003) use for their local cubic. Any fixed positive multiple satisfies both theorems: $h \rightarrow 0$ and $nh^5 \rightarrow \infty$. Write $\hat{\sigma}_{\text{ATM}}$ for the at-the-money Black–Scholes implied volatility, the median of contracts with $|K - F| \leq 0.03F$ (or the median of all finite implied volatilities if that window is empty), and take

$$h = c F \hat{\sigma}_{\text{ATM}} \sqrt{T} n^{-1/9}, \quad (16)$$

with n the number of quoted strikes and $c = 0.38$. On chains of one to several hundred strikes this is the bandwidth used for the listed prices of our estimator. Section 6.1 reports both rules.

The exponent is the one Grith et al. (2012) derive for a local cubic of a second derivative and the order Aït-Sahalia and Duarte (2003) take from Fan and Gijbels (1996). The bandwidths those papers plot are wider. Grith et al. double a cross-validated width, and plot one further widening, because the cross-validated density is wiggly, and Aït-Sahalia and Duarte report that an unconstrained local polynomial stops taking negative values only after substantial oversmoothing. Jackwerth (2004) starts from a multiple of $n^{-1/5}$. That exponent leaves nh^5 bounded, which is why it is not used. Any fixed $c > 0$ sends nh^5 to infinity. The constant $c = 0.38$ is the short end of that family.

6.1. Numerical Illustration

The data-generating process is the stochastic-volatility model of Heston (1993) with the equity parameters of Bakshi et al. (1997): $S_0 = 100$, $r = 0.05$, $q = 0$, $T = 1/4$, $\kappa = 1.15$, $\theta = 0.04$, $\sigma = 0.39$, $\rho = -0.64$, $v_0 = 0.04$. The c.f. of $\log S_T$ uses the little-trap branch of Albrecher et al. (2007). Call prices are the damped integrand of Carr and Madan (1999) with damping $\alpha = 1.5$, evaluated by the trapezoidal rule on 4096 frequencies in $(0, 250]$; puts follow by parity. The true density is Fourier inversion of the same c.f. on the same frequency grid,

$$f_{\mathbb{Q}}(K) = \frac{1}{K\pi} \int_0^\infty \text{Re}[e^{-iu \log K} \varphi(u)] du,$$

where $\varphi(u) = \mathbb{E}^{\mathbb{Q}}[e^{iu \log S_T}]$. Two chains are used: a dense grid of $m = 256$ equally spaced strikes on $[30, 220]$ (spacing $\delta = 0.745$) and a sparse grid of $m = 32$ equally spaced strikes on $[70, 140]$ (spacing $\delta = 2.26$). Integrated squared error

(ISE) is $\int_{60}^{150} (\hat{f}_{\mathbb{Q}} - f_{\mathbb{Q}})^2 dK$, evaluated by the trapezoidal rule on 401 equally spaced points. Table 1 reports ISE for the three nested estimators at the oracle h^* , at the mesh rule (15), and at (16). The oracle minimizes the integral on a geometric grid in h .

Table 1: Integrated squared error on exact Heston quotes for the cubic-spline estimator. Mesh h is (15). The last pair is (16). Quoted spline uses only the observed strikes. Tails adds the wings of Section 4.1. Tails+Thrice applies (13).

Chain	Estimator	h^*	Oracle ISE	Mesh h	Mesh ISE	$h_{1/9}$	$h_{1/9}$ ISE
Dense	Quoted spline	0.15	9.76×10^{-10}	0.89	7.36×10^{-7}	2.00	1.70×10^{-5}
Dense	Tails	0.15	9.76×10^{-10}	0.89	7.36×10^{-7}	2.00	1.70×10^{-5}
Dense	Tails+Thrice	0.63	1.16×10^{-12}	0.89	7.12×10^{-12}	2.00	2.50×10^{-8}
Sparse	Quoted spline	16.19	1.80×10^{-2}	2.71	1.04×10^{-1}	2.50	1.12×10^{-1}
Sparse	Tails	0.45	1.13×10^{-6}	2.71	5.29×10^{-5}	2.50	3.98×10^{-5}
Sparse	Tails+Thrice	2.18	9.19×10^{-7}	2.71	1.14×10^{-6}	2.50	9.98×10^{-7}

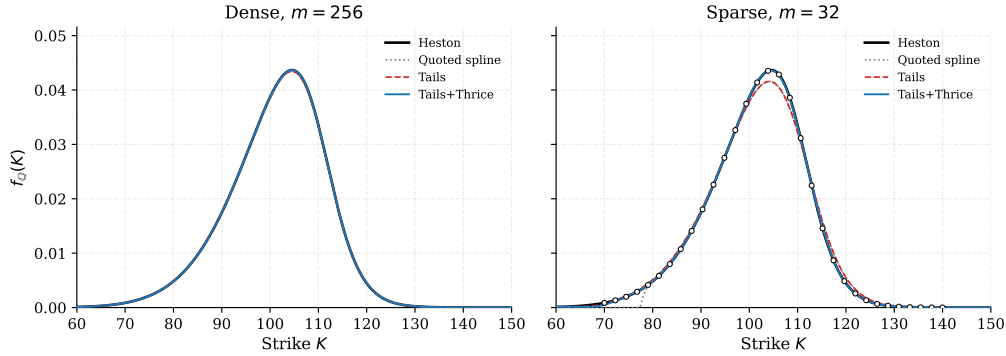


Figure 1: Heston density of S_T on $[60, 150]$. The three estimators of Table 1, each at the mesh bandwidth (15). Sparse quoted strikes are marked on the true density. Curves are truncated at zero, and the vertical scale is that of the true density.

On the dense grid the first strike lies left of the window and the last call is already zero, so the wings do not move the error. The spline without thricing has mesh integrated squared error 7.36×10^{-7} . Thricing cuts that to 7.12×10^{-12} , against an oracle 1.16×10^{-12} . Equation (16) is wider ($h = 2.00$) and the thriced error is 2.50×10^{-8} . On the sparse grid the quoted spline without tails is useless: mesh integrated squared error 0.104. The left line (10) brings the plain mesh error to 5.29×10^{-5} . Thricing then reaches 1.14×10^{-6} at the mesh and 9.98×10^{-7} at

(16) ($h = 2.50$), against an oracle 9.19×10^{-7} . Figure 1 plots the three estimators at the mesh bandwidth.

Table 2 repeats the design for the variance-gamma process of Madan et al. (1998). The process parameters are those of Carr and Madan (1999): $\sigma = 0.12$, $\nu = 0.2$, $\theta = -0.14$. Spot, rate, dividend yield, and maturity are as above, as are the two strike grids and the ISE window. Variance gamma is a pure-jump Lévy process. The true density of $\log S_T$ is a modified Bessel function (Madan et al., 1998); the density of S_T is that formula times $1/K$. Calls are the discounted payoff of that density, evaluated by the trapezoidal rule.

Table 2: Integrated squared error on exact variance-gamma quotes for the cubic-spline estimator. Columns as in Table 1.

Chain	Estimator	h^*	Oracle ISE	Mesh h	Mesh ISE	$h_{1/9}$	$h_{1/9}$ ISE
Dense	Quoted spline	0.17	5.12×10^{-7}	0.89	5.08×10^{-5}	1.28	1.55×10^{-4}
Dense	Tails	0.17	5.12×10^{-7}	0.89	5.08×10^{-5}	1.28	1.55×10^{-4}
Dense	Tails+Thrice	0.46	4.04×10^{-7}	0.89	4.90×10^{-6}	1.28	1.95×10^{-5}
Sparse	Quoted spline	14.03	3.27×10^{-2}	2.71	1.05×10^{-1}	1.59	1.78×10^{-1}
Sparse	Tails	0.45	3.97×10^{-5}	2.71	1.25×10^{-3}	1.59	3.04×10^{-4}
Sparse	Tails+Thrice	0.92	3.13×10^{-5}	2.71	3.00×10^{-4}	1.59	5.67×10^{-5}

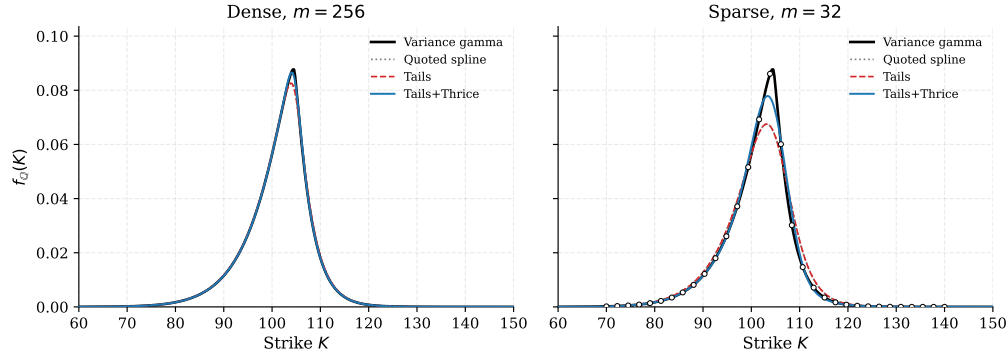


Figure 2: Variance-gamma density of S_T on $[60, 150]$. The three estimators of Table 2, each at the mesh bandwidth (15). Sparse quoted strikes are marked on the true density. Curves are truncated at zero, and the vertical scale is that of the true density.

On the dense chain the wings again do not move the digits. Thricing cuts the mesh error from 5.08×10^{-5} to 4.90×10^{-6} , against an oracle 4.04×10^{-7} . At (16) the thriced error is 1.95×10^{-5} ($h = 1.28$). The density of S_T is more peaked

than under Heston, and the absolute errors are larger. On the sparse chain the quoted spline without tails has mesh error 0.105. Tails bring the plain mesh error to 1.25×10^{-3} , and thricing lowers it to 3.00×10^{-4} at the mesh and 5.67×10^{-5} at (16) ($h = 1.59$), against an oracle 3.13×10^{-5} . Here the mesh bandwidth $h = 2.71$ is the wider of the two. Figure 2 plots the three estimators at the mesh bandwidth.

Table 3 scores the estimator of Section 4.3 against the methods of Section 7 on the same four chains. Each fit starts from the decreasing-convex projection. On these exact quotes the penalty is zero, so the row is the thriced spline at (16). The mesh bandwidth in Table 1 is smaller on dense Heston, 7.12×10^{-12} , and on dense variance gamma, 4.90×10^{-6} . The cross-validated implied-volatility cubic is the smallest error on sparse Heston, 5.66×10^{-7} , against 9.98×10^{-7} for the estimator. On sparse variance gamma the estimator has error 5.67×10^{-5} , below that cubic's 2.53×10^{-4} . The Appendix A bandwidth of Aït-Sahalia and Lo and the plug-in local linear of Aït-Sahalia and Duarte are larger on every chain.

Table 3: Integrated squared error on $[60, 150]$ for the estimator of Section 4.3 and the methods of Section 7. Ours is thriced at (16); on these exact quotes the penalty is zero. Each other row uses that estimator's bandwidth from Section 7. Grith et al. is shown at the leave-one-out width and at twice that width.

Estimator	Heston dense	Heston sparse	VG dense	VG sparse
Ours	2.50×10^{-8}	9.98×10^{-7}	1.95×10^{-5}	5.67×10^{-5}
PCA	8.55×10^{-8}	3.39×10^{-4}	2.05×10^{-4}	9.29×10^{-3}
Grith et al., IV, 2×	9.27×10^{-8}	4.34×10^{-6}	1.15×10^{-4}	7.03×10^{-4}
Grith et al., IV, CV	6.24×10^{-9}	5.66×10^{-7}	2.16×10^{-5}	2.53×10^{-4}
Aït-Sahalia and Duarte	6.13×10^{-4}	2.32×10^{-5}	6.47×10^{-3}	1.13×10^{-3}
Grith et al., call, 2×	1.14×10^{-5}	6.32×10^{-4}	4.45×10^{-4}	6.25×10^{-3}
Grith et al., call, CV	7.52×10^{-7}	5.59×10^{-5}	5.39×10^{-5}	1.36×10^{-3}
Aït-Sahalia and Lo	1.11×10^{-3}	1.09×10^{-3}	1.05×10^{-2}	6.78×10^{-3}

The same designs are repeated with noise in the quotes. Black–Scholes implied volatility receives independent $N(0, 0.01^2)$ noise. Values outside $[0.02, 2.5]$ are clipped, and the call is rebuilt. Each replication starts from the projection of those calls, and the penalty of Section 4.3 is chosen on that draw. Table 4 is that comparison. Ours is the smoothing spline. The same spline with the penalty switched off, the interpolant at $c = 0$, has mean integrated squared error 5.93×10^{-3} on the dense chain and 5.79×10^{-3} on the sparse chain. With the penalty chosen as in Section 4.3, the means are 2.22×10^{-4} and 2.58×10^{-4} . The positive convolution is 4.36×10^{-4} and 3.61×10^{-4} . Every other row in the table is larger.

Table 4: Mean integrated squared error on $[60, 150]$ for noisy Heston quotes. Implied volatility is perturbed by $N(0, 0.01^2)$ noise. Thirty replications. Ours is the smoothing spline of Section 4.3, at the bandwidth (16), with the penalty chosen on each draw. The interpolant is that spline at $c = 0$. Each other row uses that estimator’s bandwidth from Section 7.

Estimator	Dense	Sparse
Ours	2.22×10^{-4}	2.58×10^{-4}
Interpolant, $c = 0$	5.93×10^{-3}	5.79×10^{-3}
PCA	4.36×10^{-4}	3.61×10^{-4}
Grith et al., IV, $2\times$	2.56×10^{-3}	1.20×10^{-1}
Grith et al., IV, CV	1.56×10^{-2}	3.55×10^{-2}
Ait-Sahalia and Duarte	5.95×10^{-4}	2.99×10^{-3}
Grith et al., call, $2\times$	2.17×10^{-3}	1.43×10^{-3}
Grith et al., call, CV	1.54×10^{-2}	1.08×10^{-3}
Ait-Sahalia and Lo	1.61×10^{-3}	1.11×10^{-3}

7. Empirical Applications

7.1. Data and Methodology

The listed comparison uses European, AM-settled index options from Cboe’s delayed-quotes feed (`cdn.cboe.com`). Each file is dated 23 September 2026. The S&P 500 (SPX) file is 03:56 Eastern (30,470 listed contracts across 58 expiries); the Nasdaq-100 (NDX) file is 03:44 Eastern (16,492 contracts across 46 expiries); the Russell 2000 (RUT) file is 03:44 Eastern (10,790 contracts across 30 expiries). Only the standard monthly roots SPX, NDX, and RUT are kept, not weekly SPXW, NDXW, or RUTW. Time to expiry is $T = d/365.25$ with d the calendar day count from the snapshot date. The discount rate is $r = 0.04$.

A quote is retained if the bid is positive, the ask is at least the bid, the mid is at least 0.25, and the spread is at most $\max(15, 0.002S_0, 0.5 \times \text{mid})$. The 15-point term is the tolerance on the S&P 500. On the Nasdaq-100 a fixed 15-point cap deletes puts whose spreads are a dollar or two wider and leaves a gap of several thousand points in the strike list. Among remaining put–call pairs, the forward is the median of $K + e^{rT}(C - P)$ over strikes with $|K - K_{\text{atm}}| \leq 50$ on SPX, where K_{atm} is the pair closest to the cash index; on NDX and RUT the band is 0.65% of the spot. Dividend yield is $q = r - \log(F/S_0)/T$. The working slice is out of the money (OTM): the put mid if $K \leq F$, the call mid if $K \geq F$, the missing side filled by parity. Puts with $K < 0.7F$ and mid below 0.50 are then dropped; out-of-the-money calls with mid at least 0.25 are kept. Table 5 records the filter counts. In (9) and (13), $\tilde{C} = C/(S_0 e^{-qT})$ and $F = S_0 e^{(r-q)T}$.

The December 2026 S&P 500 AM quarterly ($T = 0.235$, $S_0 = 7764.64$, $F = 7836$, $q = 0.11\%$) appears in Figure 3. After the screens it has $n = 444$

strikes on $[0.36F, 1.30F]$, of which 75% are puts. Near the money the strike mesh is 5 (0.06% of F); it is 5–25 on the put wing and up to 200 beyond $1.2F$. Median out-of-the-money mids are 39 for puts and 62 for calls. Implied volatility $\hat{\sigma}_{\text{ATM}} = 0.134$ is the median Black–Scholes implied volatility of contracts with $|K - F| \leq 0.03F$. The linear call tail of Section 4.1 is appended ($C_m = 0.25$ by the mid floor of the filter, $n_R = 80$ from (11)). The bandwidth (16) uses the 444 quoted strikes and equals $h = 98.6$. The March 2027 AM quarterly ($T = 0.485$, $F = 7918$, $q = -0.04\%$, $\hat{\sigma}_{\text{ATM}} = 0.144$) filters to $n = 221$ strikes on $[0.20F, 1.49F]$, with a 25-point mesh at the money; (11) gives $n_R = 41$, and (16) gives $h = 165.9$.

The Nasdaq-100 December 2026 AM quarterly ($T = 0.235$, $S_0 = 30732$, $F = 31025$, $q = -0.03\%$, $\hat{\sigma}_{\text{ATM}} = 0.201$) is built with the same screens. It filters to $n = 312$ strikes on $[0.29F, 1.39F]$, 83% puts; the at-the-money mesh is 25, and gaps of 1000 appear beyond $1.2F$. The bandwidth (16) is $h = 606.0$ ($n = 312$; the right wing has $n_R = 72$).

The Russell 2000 December 2026 AM quarterly ($T = 0.235$, $S_0 = 2889.92$, $F = 2912$, $q = 0.80\%$, $\hat{\sigma}_{\text{ATM}} = 0.183$) filters to $n = 99$ strikes on $[0.45F, 1.32F]$, 59% puts; the at-the-money mesh is 10. The last two calls are both 0.25, so the wing slope is the previous negative step. Equation (11) gives $n_R = 40$. The bandwidth (16) is $h = 59.0$.

The quoted calls are replaced by the nearest vector in Euclidean distance that is decreasing and convex in strike. Consecutive slopes lie in $[-e^{-rT}, 0]$, and each coordinate is at least the intrinsic value $e^{-rT} \max(F - K_i, 0)$. This is the unpenalized projection of Yatchew and Härdle and also the first step of Aït-Sahalia and Duarte. Puts of the projected call follow by parity, and both premiums are floored at zero.

Out-of-the-money RMSE uses the quoted mid of the put for $K \leq F$ and that of the call for $K \geq F$. Hold-out is the average of two even/odd splits (train on even strikes and score odd, then the reverse), with h recomputed from the training n and the projection recomputed from the training strikes only.

Table 5: Construction of the working slices. “At expiry” counts listed puts and calls on that root and maturity with a positive bid and ask at least the bid. “Screens” applies the mid and spread filters. “Working OTM” is the slice used by the estimators.

Slice	At expiry	Screens	Unique K	Working OTM	Puts
SPX 18 Dec 2026	927	922	473	444	335
SPX 19 Mar 2027	453	451	228	221	165
NDX 18 Dec 2026	631	631	319	312	259
RUT 18 Dec 2026	218	218	115	99	58

7.2. Estimators

Ours. Start from the projected calls of Section 7.1. Set $P_i = 1 - C_i/(S_0 e^{-qT})$. Pass a natural cubic spline through those values, with the tail knots of Section 4.1, and invert at (16) with thricing. The listed prices are that interpolant, floored at zero, with puts by parity.

Aït-Sahalia and Duarte. Start from the projected calls of Section 7.1. The price is a Gaussian local linear regression of those values. The bandwidth is the plug-in in their Section 3.5, taken from Fan and Gijbels (1996): a global polynomial of order four supplies the pilot curvature, the weight equals one within 1.5 standard deviations of the mean strike, the Gaussian constant is 0.776, and the rate is $n^{-1/5}$. The widths are $h = 140.9, 256.1, 715.3$, and 68.8 . The density is e^{rT} times the strike derivative of the local slope, scaled so that the mass on $[\max(50, 0.2F), 2.4F]$ equals one and shifted so that the mean equals the forward. Their May 1999 figure uses the bandwidth that minimised integrated squared error in a Monte Carlo sample of 25 strikes. That optimum is not stated as a formula, and the listed row uses the plug-in. The curves have 2, 1, 4, and 4 peaks. Prices in the table are the local-linear calls, floored, with puts by parity.

Bondarenko PCA. Start from the projected calls of Section 7.1. The density is the convolution of a Gaussian kernel on the spot with a discrete probability. The nodes are equally spaced from the first quoted strike to the last, and the spacing is the bandwidth, with the count kept between 12 and 61. The weights are non-negative, sum to one, and have mean F . Each node prices a call by the Bachelier formula. The bandwidth minimises the even/odd out-of-the-money error. The selected widths are $h = 188.0, 364.1, 400.0$, and 171.3 . The NDX curve has four peaks. Prices in Table 6 are those calls, floored, with puts by parity.

Aït-Sahalia and Lo. Start from the projected calls of Section 7.1. On one maturity the futures price and the expiry are constant, and the semiparametric regression reduces to Nadaraya–Watson of Black–Scholes implied volatility on moneyness K/F . The kernel on that regressor, for the second strike derivative, is the order-2 Gaussian. With four continuous derivatives and one regressor, the bandwidth in their Appendix A is $c s(K/F) n^{-1/9}$, and $c = c_0/\ln n$. Table II of that paper gives $c = 1.260$ at $n = 14,431$, which fixes c_0 . The bandwidths are $h = 0.171, 0.303, 0.247$, and 0.315 in units of moneyness. Prices are the Black–Scholes calls at the smoothed volatility, floored, with puts by parity. The density is the strike derivative of that call. Each curve has one peak and mass 1.0.

Grith et al.. Start from the projected calls of Section 7.1. Both objects are a local cubic with the quartic kernel, and h is the half-width of that kernel. The bandwidth minimises the leave-one-out squared error of the local cubic, the cross-validation they compute and then set aside because the density is wiggly. The width in their comparison figure is twice that value: on the DAX chain, 114 against 228. The call row divides the call by the spot, reads the density as e^{rT} times the scaled second derivative, and prices from the local level. Cross-validation selects $h = 52.2, 317.9, 498.7$, and 110.7 , so the doubled widths are $104.4, 635.7, 997.4$, and 221.5 . The implied-volatility row is a local cubic of Black–Scholes implied volatility against the strike, with Black–Scholes prices at the local level and a density that uses the local slope and curvature. Its cross-validated widths are $h = 88.9, 504.8, 498.7$, and 204.0 , and the doubled widths are $177.9, 1009.6, 997.4$, and 407.9 . Their DAX figure for this row shows one widened curve, at 372, and does not print the cross-validated width, so the same doubling is used. The doubled call cubic is drawn on the left of Figure 3. Figure 4 draws the cross-validated call cubic and both implied-volatility widths, together with Ait-Sahalia and Lo. Prices are floored, with puts by parity.

Hold-out refits the projection and every tuned method on the training strikes only, including the choice of its tuning constant, and scores the other half against the quoted mids. The two directions are averaged.

On SPX December the quoted calls sit 0.022 from the decreasing-convex cone, so a fit that stays inside the cone cannot undercut that error. The local linear of Ait-Sahalia and Duarte at the plug-in bandwidth is 4.272, 7.837, 16.023, and 2.224. With a Gaussian kernel on the spot and a width chosen by pricing error, PCA out-of-the-money error is 0.051 on SPX December, 0.066 on SPX March, 1.151 on NDX December, and 0.058 on RUT December. The NDX density has four peaks.

The estimator of this paper is the smoothing spline of P in (14), with tails and thricing, at (16). The pricing error is 0.036, 0.056, 1.065, and 0.026. The hold-out error is 0.041, 0.066, 1.109, and 0.031. Each of the four curves has one peak on $[0.55F, 1.40F]$ and mass 1.0. On SPX December, SPX March, and RUT the selected penalty leaves the prices on the interpolant. On NDX the penalty is 1.44×10^{-6} . At the mesh bandwidth the same spline has out-of-the-money error 0.037, 0.066, 1.441, and 0.029, and hold-out 1.419, 0.054, 1.291, and 0.536. Each mesh density still has one peak, and mass and total variation print as 1.0.

Ait-Sahalia and Lo, at the Appendix A bandwidth, has error 31.225, 75.231, 179.751, and 12.871, with one peak on each chain and mass 1.0. The cross-validated call cubic of Grith et al. prices at 0.029, 0.042, 0.867, and 0.022, and the doubled width at 0.031, 0.152, 0.946, and 0.056. The cross-validated implied-

Table 6: Out-of-the-money repricing. Every fit starts from the projection in Section 7.1 and is scored on the quoted mids. Ours is the cubic-spline inversion at (16) ($h = 98.6, 165.9, 606.0, 59.0$). Ait-Sahalia and Duarte is the constrained local linear at the Fan–Gijbels plug-in; the mass is that density after the unit-mass scaling. Ait-Sahalia and Lo uses the Appendix A bandwidth, in moneyiness units $h = 0.171, 0.303, 0.247$, and 0.315 . Each Grith et al. object is shown at the leave-one-out width and at twice that width. The mass is the density of that same fit. PCA is priced from the Gaussian convolution on the spot. OTM is put RMSE for $K \leq F$ and call RMSE for $K \geq F$, premiums floored at zero. Mass is $\int \max(\hat{q}, 0)$ on $[\max(50, 0.2F), 2.4F]$. TV is the total variation of $\max(\hat{q}, 0)$ on $[0.55F, 1.40F]$, divided by twice the maximum; a unimodal curve scores 1.0. Hold-out refits the projection and the tuning on each half. Average is the equal-weight mean of the four chains. The bold method name is the spline at (16). The mesh row uses the same smoothing spline at (15), with widths $h = 12.0, 30.0, 120.0$, and 12.0 .

Chain	Method	RMSE OTM	Puts	Calls	Hold-out	Mass	TV
SPX Dec	Ours	0.036	0.034	0.041	0.041	1.0	1.0
SPX Dec	Ours, mesh	0.037	0.033	0.048	1.419	1.0	1.0
SPX Dec	PCA	0.051	0.053	0.048	0.055	1.0	1.0
SPX Dec	Grith et al., IV, 2×	0.032	0.031	0.037	0.055	1.0	1.1
SPX Dec	Grith et al., IV, CV	0.031	0.030	0.035	0.074	1.0	1.3
SPX Dec	Ait-Sahalia and Duarte	4.272	3.263	6.450	5.385	1.0	1.0
SPX Dec	Grith et al., call, 2×	0.031	0.030	0.036	9.964	1.0	1.1
SPX Dec	Grith et al., call, CV	0.029	0.029	0.032	14.653	1.0	3.4
SPX Dec	Ait-Sahalia and Lo	31.225	22.162	49.619	37.922	1.0	1.0
SPX Mar	Ours	0.056	0.052	0.067	0.066	1.0	1.0
SPX Mar	Ours, mesh	0.066	0.054	0.092	0.054	1.0	1.0
SPX Mar	PCA	0.066	0.061	0.077	0.075	1.0	1.0
SPX Mar	Grith et al., IV, 2×	0.165	0.143	0.215	0.282	1.0	1.0
SPX Mar	Grith et al., IV, CV	0.051	0.049	0.057	0.091	1.0	1.0
SPX Mar	Ait-Sahalia and Duarte	7.837	5.633	12.202	9.965	1.0	1.0
SPX Mar	Grith et al., call, 2×	0.152	0.107	0.240	1.339	1.0	1.0
SPX Mar	Grith et al., call, CV	0.042	0.040	0.048	18.635	1.0	1.0
SPX Mar	Ait-Sahalia and Lo	75.231	53.202	118.303	89.227	1.0	1.0
NDX Dec	Ours	1.065	1.113	0.792	1.109	1.0	1.0
NDX Dec	Ours, mesh	1.441	1.219	2.228	1.291	1.0	1.0
NDX Dec	PCA	1.151	1.117	1.304	1.324	1.0	1.3
NDX Dec	Grith et al., IV, 2×	0.946	0.991	0.686	1.236	1.0	1.2
NDX Dec	Grith et al., IV, CV	0.867	0.921	0.524	1.248	1.0	3.0
NDX Dec	Ait-Sahalia and Duarte	16.023	14.909	20.619	19.189	1.0	1.1
NDX Dec	Grith et al., call, 2×	0.946	0.993	0.677	40.791	1.0	1.1
NDX Dec	Grith et al., call, CV	0.867	0.922	0.523	79.411	0.9	2.8
NDX Dec	Ait-Sahalia and Lo	179.751	157.159	263.641	241.766	1.0	1.0
RUT Dec	Ours	0.026	0.027	0.024	0.031	1.0	1.0
RUT Dec	Ours, mesh	0.029	0.027	0.031	0.536	1.0	1.0
RUT Dec	PCA	0.058	0.066	0.043	0.067	1.0	1.0
RUT Dec	Grith et al., IV, 2×	0.042	0.047	0.033	0.116	1.3	1.0
RUT Dec	Grith et al., IV, CV	0.026	0.027	0.025	0.039	1.0	1.1
RUT Dec	Ait-Sahalia and Duarte	2.224	1.968	2.544	2.834	1.0	1.1
RUT Dec	Grith et al., call, 2×	0.056	0.051	0.063	0.432	1.0	1.0
RUT Dec	Grith et al., call, CV	0.022	0.022	0.023	7.026	1.0	1.1
RUT Dec	Ait-Sahalia and Lo	12.871	9.304	16.660	15.912	1.0	1.0
Average	Ours	0.296	0.306	0.231	0.312	1.0	1.0
Average	Ours, mesh	0.393	0.333	0.600	0.825	1.0	1.0
Average	PCA	0.331	0.324	0.368	0.380	1.0	1.1
Average	Grith et al., IV, 2×	0.296	0.303	0.243	0.422	1.1	1.1
Average	Grith et al., IV, CV	0.244	0.257	0.160	0.363	1.0	1.6
Average	Ait-Sahalia and Duarte	7.589	6.443	10.454	9.343	1.0	1.0
Average	Grith et al., call, 2×	0.297	0.295	0.254	13.131	1.0	1.1
Average	Grith et al., call, CV	0.240	0.253	0.156	29.932	1.0	2.1
Average	Ait-Sahalia and Lo	74.769	60.456	112.056	96.207	1.0	1.0

volatility cubic prices at 0.031, 0.051, 0.867, and 0.026, and the doubled width at 0.032, 0.165, 0.946, and 0.042. The call cubic's hold-out is 14.653, 18.635, 79.411, and 7.026 at the cross-validated width, and 9.964, 1.339, 40.791, and 0.432 at twice that width. Those errors sit in the far wing, where a one-sided local cubic on the held-out strikes misses the put.

A peak count treats a ripple and a spike as the same event, and the count depends on where the height cutoff is set. The TV column of Table 6 is the total variation of $\max(\hat{q}, 0)$ on the strike window of Figure 3, divided by twice the highest value on that window. A unimodal curve climbs once and descends once, so the ratio equals one, apart from whatever mass remains at the two edges. An extra oscillation of height a adds $2a$ to the numerator and so adds a divided by the height of the mode. The bold row scores 1.0. At the cross-validated width the call cubic scores 3.4, 1.0, 2.8, and 1.1, and at twice that width 1.1, 1.0, 1.1, and 1.0. The implied-volatility cubic scores 1.3, 1.0, 3.0, and 1.1 at the cross-validated width, and 1.1, 1.0, 1.2, and 1.0 at twice that width. Aït-Sahalia and Lo scores 1.0 on each chain. Aït-Sahalia and Duarte scores 1.0, 1.0, 1.1, and 1.1. The doubled call cubic and that local linear are on the left of Figure 3. Figure 4 draws Aït-Sahalia and Lo, the cross-validated call cubic, and both implied-volatility widths on that same vertical scale.

The average out-of-the-money hold-out error is 0.312 for the estimator of this paper, 0.380 for PCA, 9.343 for Aït-Sahalia and Duarte, and 96.207 for Aït-Sahalia and Lo. The doubled and cross-validated implied-volatility cubics of Grith et al. average 0.422 and 0.363, and the call cubics 13.131 and 29.932.

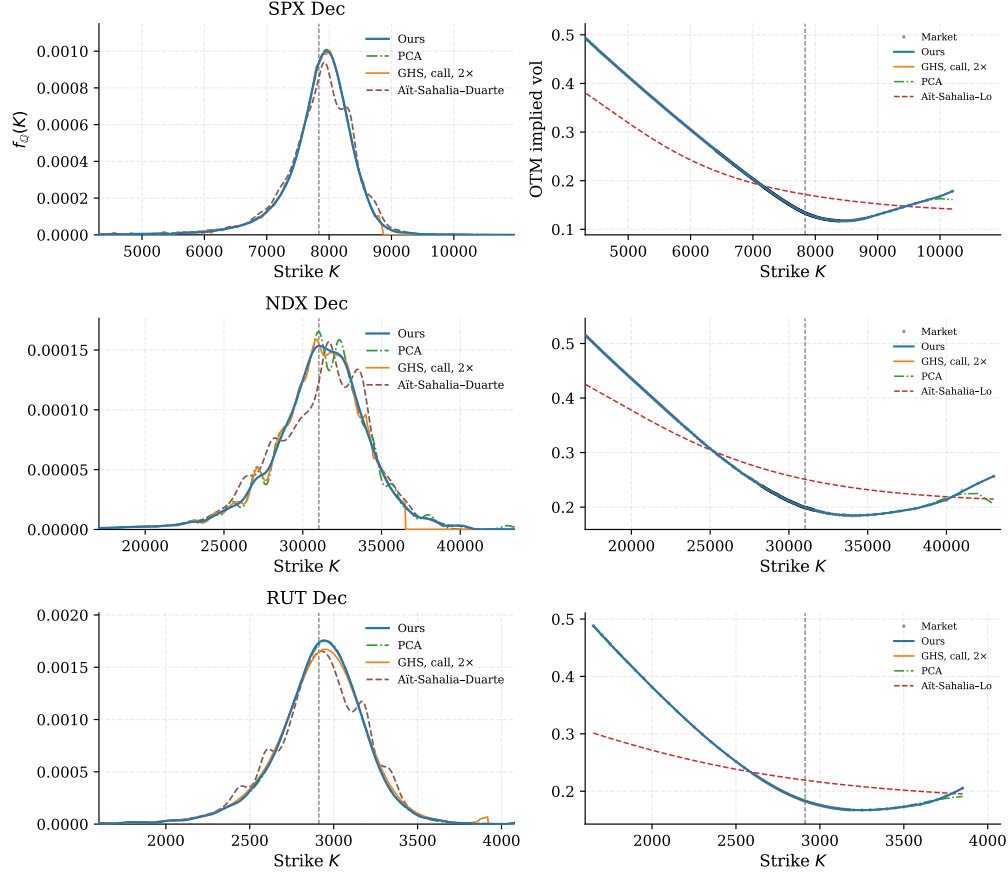


Figure 3: SPX, NDX, and RUT, 18 December 2026, snapshot 23 September 2026. Each curve is fit to the projected call of Section 7.1. Left: $\max(\hat{f}_Q, 0)$ for the smoothing spline at (16), the Gaussian convolution of Bondarenko, the call cubic of Grith et al. at twice its cross-validated bandwidth, and the constrained local linear of Ait-Sahalia and Duarte at the Fan–Gijbels plug-in. Right: out-of-the-money implied volatility for the smoothing spline, the Gaussian convolution, that call cubic, and Ait-Sahalia and Lo at the Appendix A bandwidth. The dashed line is the forward. Ait-Sahalia and Lo, the cross-validated call cubic, and the implied-volatility cubics are in Figure 4.

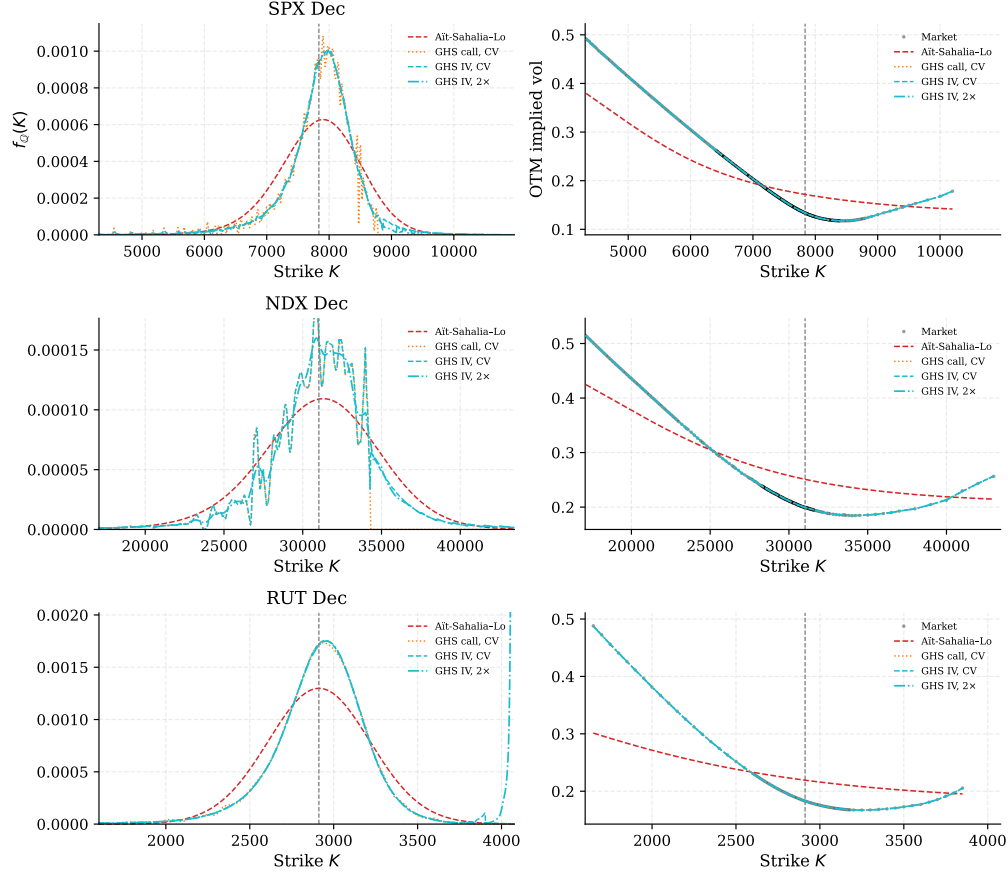


Figure 4: Further kernels on the chains of Figure 3. Ait-Sahalia and Lo uses the Appendix A moneyness bandwidth. The call cubic of Grith et al. is at the leave-one-out width, and the implied-volatility cubic is drawn at that width and at twice that width. Left: $\max(\hat{f}_Q, 0)$, on the strike window and vertical scale of the left panel of Figure 3, with curves truncated there. Right: out-of-the-money implied volatility of the same fits. The dashed line is the forward.

8. Conclusion

The estimator reads the risk-neutral density off the empirical measure of a single maturity. Calls are rescaled to a complementary distribution, a smoothing spline fits that measure, the unquoted tails are filled, and a thrice Gaussian convolution of the spline is the density and the call. The penalty is zero when the thrice density is already unimodal. On exact quotes that is the selected value, so the integrated squared error is the thrice interpolant at (16). The mesh bandwidth in Table 1 is smaller on dense Heston and on dense variance gamma. On sparse

Heston the cross-validated implied-volatility cubic of Grith et al. is smaller. With one volatility point of noise in implied volatility, the smoothing spline has mean integrated squared error 2.22×10^{-4} and 2.58×10^{-4} , below the positive convolution. On the four listed chains the density has one peak and mass 1.0, and the out-of-the-money error is below that convolution on each chain. The Appendix A bandwidth of the Nadaraya–Watson smile of Aït-Sahalia and Lo leaves one peak and mass 1.0, with error 31.225, 75.231, 179.751, and 12.871. The constrained local linear of Aït-Sahalia and Duarte has error 4.272, 7.837, 16.023, and 2.224. The leave-one-out call cubic of Grith et al. reprices in sample, and twice that width is the curve in their comparison figure; neither width holds out on the far wing.

Code and Data Availability

A Python implementation of (13) and a replication script are available at <https://github.com/s-broda/RND>.

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